



CODEKSHETRA HACKATHON

Round 1: Idea Proposal Document

Track: Software Edition Only

Team Information

No.	Member Name	Institute Name	Role
1	Aditya Dileep Shintre	DKTE	Team Leader
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Team Size Requirement: Minimum 2 and Maximum 4 Members.

Team Name: THE HOISTERS

Project Name: Claim Sheild – Smart Medical Coding and Denial Prediction System

Selected Theme: Health Tech

ClaimShield AI – Smart Medical Coding & Denial Prediction System

1. Abstract

Healthcare systems worldwide lose billions annually due to inefficient medical coding and high insurance claim denial rates.

Manual claim processing is slow, error-prone, and reactive, resulting in revenue leakage, delayed reimbursements, and administrative burden.

ClaimShield AI is an AI-powered Revenue Cycle Intelligence Engine that automates medical code extraction from clinical notes and predicts insurance claim denial risk before submission. Using NLP-based diagnosis extraction and machine learning-based denial risk modeling, the system transforms unstructured clinical documentation into structured claim-ready outputs while proactively flagging high-risk claims.

Unlike traditional billing systems that detect errors after claim rejection, ClaimShield AI introduces a predictive pre-adjudication layer that reduces denial probability, improves financial efficiency, and accelerates reimbursements.

2. Problem Statement

Healthcare providers face high insurance claim denial rates (5–20%), leading to revenue loss, increased rework costs, delayed cash flow, and administrative burden. Manual medical coding from unstructured clinical notes is time-consuming and error-prone.

There is a need for an intelligent system that automates clinical-to-code translation, predicts claim denial probability before submission, and assists billing teams in reducing financial risk proactively.

3. Market Analysis – Pros and Cons

Market Need:

- Healthcare administrative waste costs billions annually.
- Increasing insurance policy complexity.
- Rising denial rates due to documentation errors.

Pros:

- Strong demand from hospitals and RCM companies.

- Scalable across healthcare networks.
- High ROI due to reduced denial rework costs.

Cons:

- Requires regulatory compliance in real deployment.
- Integration with existing hospital systems may be complex.
- Model accuracy depends on quality of training data.

4. Target Users

Primary Users:

- Medical Billing Specialists
- Revenue Cycle Management (RCM) Teams
- Hospital Billing Departments

Secondary Users:

- Hospital Revenue Managers
- Healthcare Financial Analysts

5. Existing Solutions

Current solutions include manual coding workflows, static rule-based claim scrubbers, and traditional RCM software with limited AI intelligence.

Limitations:

- Reactive denial handling
- Limited predictive capability
- Lack of intelligent pattern learning

6. Competitive Advantage

- AI-driven medical code extraction from unstructured notes
- Predictive denial risk scoring before claim submission
- Intelligent explanation system for risk factors
- Scalable architecture for integration

7. Market Impact

- Reduce claim denial rates (prototype simulation: 40–60% potential reduction)
- Reduce Accounts Receivable cycle time
- Lower administrative rework costs
- Improve hospital revenue efficiency

8. Proposed Solution

Modules:

1. Clinical NLP Engine – Extracts diagnoses and maps to ICD-10 codes.
2. Claim Structuring Module – Combines ICD, CPT, payer, and charges.
3. Denial Prediction Engine – Uses ML (XGBoost) to predict denial probability and assign risk level.

System Flow:

Clinical Note Upload → Diagnosis Extraction → ICD Mapping → User Adds CPT & Payer → Denial Prediction → Risk Score + Claim Summary Report

9. Technology Stack

Component	Technology Used
Frontend	Streamlit
Backend	FastAPI / Python
Database	SQLite
Cloud / Deployment	Cloud / Localhost
Other Tools / APIs	Scikit-learn, XGBoost, HuggingFace Transformers

Innovativeness

- Predictive denial modeling before submission
- AI-based clinical-to-code automation
- Risk-based claim routing
- Intelligent explanation layer

Show Stoppers and Risk Analysis

Technical Risks:

- Limited dataset for training
- Model accuracy variation

Regulatory Risks:

- Requires compliance in real-world deployment

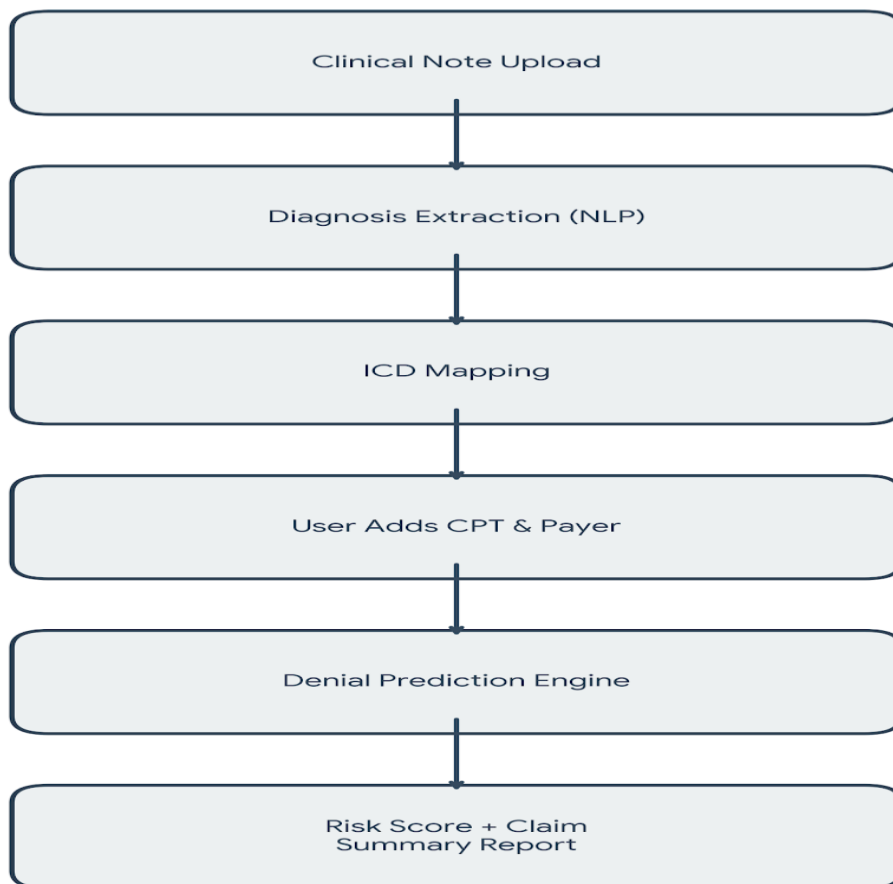
Scalability Challenges:

- Integration with EHR systems
- Multi-payer policy variations

Use Case / System Flow Diagram

System Workflow Diagram:

Workflow Diagram



Prototype / Wireframes

Below are the UI Screenshots of ClaimShield AI:

Link - <https://github.com/aditya764k/Codeshetra2026>

Upload Clinical Note

ClaimShield AI

Smart Medical Coding & Denial Prediction System

Upload a patient's clinical note, and our AI will extract relevant ICD codes and predict denial risk for insurance claims.

1. Upload Clinical Note

Drop clinical note file here > or click to upload

clinical_note_example.txt

Chief Complaint:
Intermittent dizziness routine follow up.

History of present illness: 111 were add nali with history of hypertension and type 2 diabetes mellitus.

Blood pressure today: 110/80/35 mmHg.

Patient report: consistent high blood pressure reading at home.

Medications: Lisinpril 22mg daily, Metformin 1000mg twice daily.

Assessment:
Hypertension, uncontrolled; Type 2 diabetes mellitus, without complication.

Plan: Amlodipine 5mg daily, Continue Metformin at current dose, Order repeat blood tests including HbA1c and lipid panel.

Process Note

ICD Code Extraction

ClaimShield AI

Smart Medical Coding & Denial Prediction System

2. Extracted ICD Codes

Diagnosis	ICD-10 Code
Essential Hypertension	I10
Type 2 Diabetes Mellitus; Without Complications	E11.9

ICD-10 codes extracted completed successfully.
Please proceed by selecting the CPT code and payer below.

Select CPT Code:
99214 - Office Visit, Level 4

Payer:
Medicare

Predict Denial Risk

Denial Risk Dashboard

Denial Risk Dashboard

ClaimShield AI

3. Denial Risk Prediction

Low Medium High Very High

Denial Risk: 72% Model Confidence: 85%

Explanation

This claim is classified as HIGH RISK because the combination of ICD 110 Essential Hypertension (with CPT code 99214 (Office Visit, Level 4), has historicals upon a nly denial rate rate Medicare often, 23) to insufficient medical history necessary documentation.

Claim Details:

Diagnosis:	Essential Hypertension
CPT:	99214 - Office Visit, Level 4
Payer:	Medicare
Estimated Charge:	\$5.90

Claim Summary Report

Claim Summary Report

ClaimShield AI

4. Claim Ready Summary

Patient:	Demo Patient
Diagnosis:	Essential Hypertension E11.9 - Type 2 Diabetes Mellitus Without Complications
CPT Code:	99214 - Office
Payer:	Medicare
Estimated Charge:	3200

Risk Explanation

This claim is classified as HIGH RISK because the combination of ICD 110 Essential Hypertension (with the CPT code 99214 (Office Visit, Level 4), has historicals upon a nly denial rate rate with Medicare often use at insufficient medical history necessary documentation.

Download Claim Report (PDF)

Declaration

We declare that this idea is original and developed for CODEKSHETRA 2026 Hackathon. Any reused components have been properly disclosed.